DAVI CARVALHO ARAÚJO

AUTOMOTIVE ENGINEERING SPECIALIST | COMBUSTION & AFTERTREATMENT CONTROLS | TECHNICAL LEADERSHIP

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PROFESSIONAL PROFILE

Automotive controls engineer and technical leader with end-to-end experience in requirements, physics-based control design, model-based development, AUTOSAR integration, calibration support, and MIL/SIL/HIL validation. Product Owner for Combustion & Aftertreatment controls across conventional and hybrid propulsion programs, while remaining hands-on in control design, technical reviews, release content, and complex issue resolution. Recent postgraduate specialization in Automotive Engineering & Mobility focused on hybrid and electric vehicles.

PROFESSIONAL EXPERIENCE

Product Development Coordinator | Stellantis - São Paulo, SP
Sep 2024 - Present

- **Technical portfolio ownership:** Serve as Product Owner and technical leader for Combustion & Aftertreatment controls, coordinating priorities, design maturity, software documentation, validation evidence, and production-release content across multiple vehicle programs.
- **Production programs:** Support Hurricane 4 EVO (I-4 2.0L Turbo with TJI, multiple-pulse PDI, and GPF), Hurricane 6 (I-6 3.0L Bi-Turbo, standard and high-output), and Pentastar Upgrade (V6 3.6L VVL + EGR integrated into a REEV architecture).
- **Requirements and control design:** Lead definition and review of functional requirements and their translation into physics-based control strategies covering spark, knock, fuel dynamics, torque estimation, thermal management, oxygen sensing, catalyst control, and GPF functions.
- **Technical governance:** Review and approve control designs, prioritize backlog and release scope, and maintain alignment and traceability across requirements, architecture, implementation, calibration, and validation.
- **Cross-functional leadership:** Lead technical discussions and root-cause investigations across controls software, calibration, validation, BSW suppliers, and AUTOSAR system architecture teams, while remaining hands-on in design decisions and issue resolution.

Product Engineer | Bosch - Campinas, SP
Jan 2023 - Sep 2024

- **Control-system technical sponsor:** Held end-to-end technical responsibility for assigned control-system components, from requirements interpretation and control-design definition through integration, initial calibration, bench validation, and production support.
- **Model-based development:** Designed and maintained physics-based control models in MATLAB/Simulink and Stateflow, translating functional requirements into production-oriented logic for conventional and hybrid propulsion systems.
- **Architecture and embedded integration:** Defined AUTOSAR software architecture and interfaces in dSPACE SystemDesk, supported generated C-code integration, and debugged control behavior on embedded targets.
- **Verification and issue resolution:** Planned and executed MIL, SIL, PIL, and HIL verification; diagnosed software and calibration issues from bench, vehicle, and post-production testing, tracing findings to control logic, interfaces, or requirement gaps.
- **Technical delivery and Scrum:** Combined hands-on engineering with Scrum Master responsibilities, coordinating sprint planning, execution alignment, dependency management, and removal of delivery blockers within a global cross-functional team.

Young Talent Internship Program | Bosch - Campinas, SP
Sep 2022 - Jan 2023

- **Automotive foundation:** Selected for a structured development program covering ICE, hybrid, and electric propulsion systems, control fundamentals, and calibration principles.
- **Calibration exposure:** Supported engine and transmission calibration activities across automatic, manual, and continuously variable transmission applications using ETAS INCA.
- **Requirements and simulation:** Contributed to high-level functional requirements and applied MATLAB/Simulink to Model-in-the-Loop simulation and validation of control logic.
- **Engineering automation:** Developed Python and C++ scripts supporting calibration, data analysis, and dataset-management workflows.

EDUCATION

Postgraduate Specialization | Instituto Mauá de Tecnologia - São Paulo, SP

2025 - 2026

Automotive Engineering & Mobility - Hybrid and Electric Vehicles

Hybrid and electric propulsion: xHEV platforms, electric machines, HV batteries, automotive power electronics, charging infrastructure, and HV safety.

Vehicle engineering: Vehicle dynamics, brakes, suspension and steering, vehicle architecture, NVH, structural analysis, safety, quality, and reliability.

Simulation and validation: HEV/EV modeling, vehicle-dynamics simulation, dynamometer testing, laboratory activities, testing, and homologation.

B.Sc. in Mechanical Engineering | Federal University of Sergipe - São Cristóvão, SE

2018 - 2023

Academic performance: Grade: 8.5/10.0.

Featured project: Fuel Heating Control System for Automobiles - mathematical modeling and PID control in MATLAB/Simulink for pre-start and post-start Flex-Fuel operation. **Published article**

TECHNICAL EXPERTISE

CONTROLS DOMAINS	Combustion and ignition control; fuel and air-fuel ratio control; knock and pre-ignition protection; torque and combustion estimation; aftertreatment and thermal management; transient and adaptive control.
REQUIREMENTS & DESIGN	IBM DOORS Next / ELM; ASPICE; dSPACE SystemDesk; CANdelaStudio; AUTOSAR.
DEVELOPMENT	MATLAB, Simulink, Stateflow, C, C++, Python; physics-based model development and embedded software integration.
VERIFICATION	MIL, SIL, PIL, HIL; BTC EmbeddedPlatform; requirement-based verification and root-cause analysis.
INTEGRATION	TeamCity CI pipelines; Python automation; Lauterbach TRACE32 debugging and trace.
CALIBRATION & TEST	ETAS INCA, ETAS MDA, MATLAB, dSPACE ControlDesk, PROVEtech; calibration support, measurement analysis, bench and vehicle validation.

LANGUAGES

Portuguese - Native | **English** - Fluent